Yin-Kai Yu (余荫铠)

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GitHub | Google Scholar | ORCID | arXiv | ResearchGate | Bilibili | Zhihu | Xiaohongshu

My research focuses on condensed-matter theory, particularly strongly correlated systems, nonequilibrium statistical physics, phase transitions and critical phenomena, and quantum Monte Carlo methods, especially the sign problem.

I excel at analytical derivations and large-scale numerical computation. I collaborate fluently with ChatGPT and Codex across numerical studies and formal theory, and I have a strong interest in accelerating scientific discovery with AI.

Education

- **Institute of Physics, Chinese Academy of Sciences** 2024 – 2029 (expected)
  - Ph.D. Student in Theoretical Physics, University of Chinese Academy of Sciences.
  - Advisor: Prof. Zi-Xiang Li.
- **Sun Yat-sen University, School of Physics** 2020 – 2024
  - B.S. in Physics; GPA: 92/100; Rank: 2/79.
  - National Scholarship.
  - Advisor: Prof. Shuai Yin; thesis: *Research on the sign problem in non-equilibrium quantum criticality*.

Publications

Peer-reviewed (# denotes co-first authors)

[1] **Science Advances 12, eadz4856 (2026)**
Preempting fermion sign problem: Unveiling quantum criticality through nonequilibrium dynamics in imaginary time
Yin-Kai Yu, Zhi-Xuan Li, Shuai Yin, and Zi-Xiang Li.
（提出缓解费米子符号问题的通用方法；发现了新的非常规的费米子相变普适类）

[2] **Physical Review Letters 136, 086502 (2026)**
Nonequilibrium Dynamics of Dirac Quantum Criticality in Imaginary Time
Yin-Kai Yu, Zhi Zeng, Yu-Rong Shu, Zi-Xiang Li, and Shuai Yin.
（首次研究费米子相变的非平衡临界动力学）

[3] **Nature Communications 16, 6181 (2025)**
Finite-time scaling beyond the Kibble–Zurek prerequisite in Dirac systems
Zhi Zeng[#], **Yin-Kai Yu**[#], Zhi-Xuan Li[#], Zi-Xiang Li, and Shuai Yin.
（超越并推广了著名的 Kibble–Zurek 机制）

[4] **Chinese Physics Letters 43, 090710 (2026)**
Symmetric Mass Generation Transition and its Nonequilibrium Critical Dynamics in a Bilayer Honeycomb Lattice Model
Zhi-Xuan Li, **Yin-Kai Yu**, Zi-Xiang Li, and Shuai Yin.

[5] **Physical Review B 112, L060301 (2025)**
Nonequilibrium critical dynamics with emergent supersymmetry
Zhi Zeng, **Yin-Kai Yu**, Zi-Xiang Li, and Shuai Yin.

Preprints (# denotes co-first authors)

[1] arXiv:2610.31541 (2026)
Sign-problem-free Bosonic Auxiliary Field Quantum Monte Carlo
Yin-Kai Yu, Zhong-Chao Wei, and Zi-Xiang Li.
（从根源解决阻挫玻色系统的符号问题）

[2] arXiv:2608.07990 (2026)
Metallic Gross–Neveu criticality and superconductivity on the SO(3) SLAC fermion
Feng-Yu Zhang[#], **Yin-Kai Yu**[#], Zi-Xiang Li, and Shuai Yin.

[3] arXiv:2512.13546 (2025)
Magnetic order and novel quantum criticality in the strongly interacting quasicrystals
Cong Zhang, **Yin-Kai Yu**, Shao-Hang Shi, and Zi-Xiang Li.

Selected Awards

- **UCAS Merit Student** 2025
- **National Scholarship** 2023
- **National Endeavor Scholarship** 2021, 2022
- **SYSU Outstanding Student Scholarship, First Prize** 2021, 2022, 2023

Invited talks and presentations

Invited lectures

[1] **Nonequilibrium critical dynamics of strongly correlated fermions** Mar 12, 2026
Zhejiang Sci-Tech University, online (3 hours); video.

[2] **Nonequilibrium critical dynamics of strongly correlated fermions** Oct 15, 2025
Sun Yat-sen University.

Oral presentations

[1] **Before Fermi Statistics Awakens** Nov 23, 2024
Workshop on Developments and Applications of Monte Carlo Algorithms, Westlake University (video).

[2] **Nonequilibrium quantum criticality of interacting Dirac fermions** Jul 28, 2023
7th National Conference on Statistical Physics and Complex Systems (video).

Posters

[1] **Sign-problem-free Bosonic Auxiliary Field Quantum Monte Carlo** Apr 10–12, 2026
14th Workshop on Quantum Many-Body Computation.

[2] **Preempting fermion sign problem** Jan 15–16, 2026
Spring Academic Exchange Conference, Institute of Physics, CAS.

[3] **Solving SU(3) Hubbard model with sign problem** Apr 11–13, 2025
13th Workshop on Quantum Many-Body Computation.

[4] **Preempting fermion sign problem** Apr 12–14, 2024
12th Workshop on Quantum Many-Body Computation.

[5] **Nonequilibrium quantum criticality of interacting Dirac fermions** Jul 27, 2023
7th National Conference on Statistical Physics and Complex Systems.

Open source contributions

- **BAFQMC** (Bosonic Auxiliary-Field Quantum Monte Carlo): Resolves the sign problem at its root in frustrated bosonic systems, enabling large-scale, unbiased, sign-problem-free simulations even in certain no-go systems previously considered to have an intrinsic sign problem. Agent-ready simulation and model-extension workflows.
- **STQMC** (Short-Time Quantum Monte Carlo): State-of-the-art algorithm underpinning Sci. Adv. 12, eadz4856 and Phys. Rev. Lett. 136, 086502, extracting quantum critical points and exponents through short-time nonequilibrium scaling before the fermion sign problem becomes severe.
- **topology-gomoku**: Original, free, cross-platform indie game with novel Gomoku gameplay on topological surfaces and interactive 2D–3D visualization; 50,000 monthly active players across all platforms.
- **paper-wordcloud**: Scientific-paper word-cloud generator with custom masks and automated text processing; 45 GitHub stars/forks combined and widely used in the Xiaohongshu community.

Skills

- **Quantum Many-Body Physics:** Theory and numerical methods including quantum Monte Carlo, exact diagonalization, tensor networks, mean-field theory, and variational methods.
- **Generative AI, Agents, and Harnesses:** Generative-model and LLM fundamentals and architectures; agent design, tools, evaluation, and benchmarking; hands-on use of Codex and Claude Code; agent infrastructure for research and software development.
- **Engineering and HPC:** HPC for large-scale scientific workloads; Slurm and PBS job scheduling, MPI parallelization, toolchains, and environment management. Access to a cluster with 2,000+ CPU cores.
- **Languages:** Python, Fortran, Julia, C, MATLAB, Mathematica, Bash, JavaScript, HTML, CSS, Markdown, LaTeX; Chinese, English.